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Introduction

A pulsar is a highly magnetized neutron star which emits beams of electromagnetic radiation, which manifest as radio waves. The first pulsar was detected in 1967, and was found by manually combing through data, looking for repeating frequency pulses. Since then, many improvements have been made to this data analysis pipeline especially as detection technology and software engineering have progressed. As such, the goal of this research project was to create a data analysis pipeline which would automatically detect a pulsar signal, strengthen it with stacking, and fold to create a pulse profile, within the constraint of a week-long timeframe.

Methods

Prior to the start of this project, some reading was done to gain a firmer understanding of what steps must be taken to meet the objective. The trajectory of this project followed that as outlined in the CHAMPSS experiment (Andrade, 2025), with the deviation that only one dataset from one location on one day would be used.
In this way, the first step of this project was to create a raw spectral intensity plot from a 16-minute CHIME dataset in order to see the recorded frequencies, which was mostly RFI observationally. This dataset was then cropped down to about a 30-second timeframe (from 5s to 35s) picked based on a timeframe with low visible RFI spanning the frequency spectrum. A background median was then subtracted from the cropped dataset in order to initially mitigate the effects of RFI, as can be seen below.

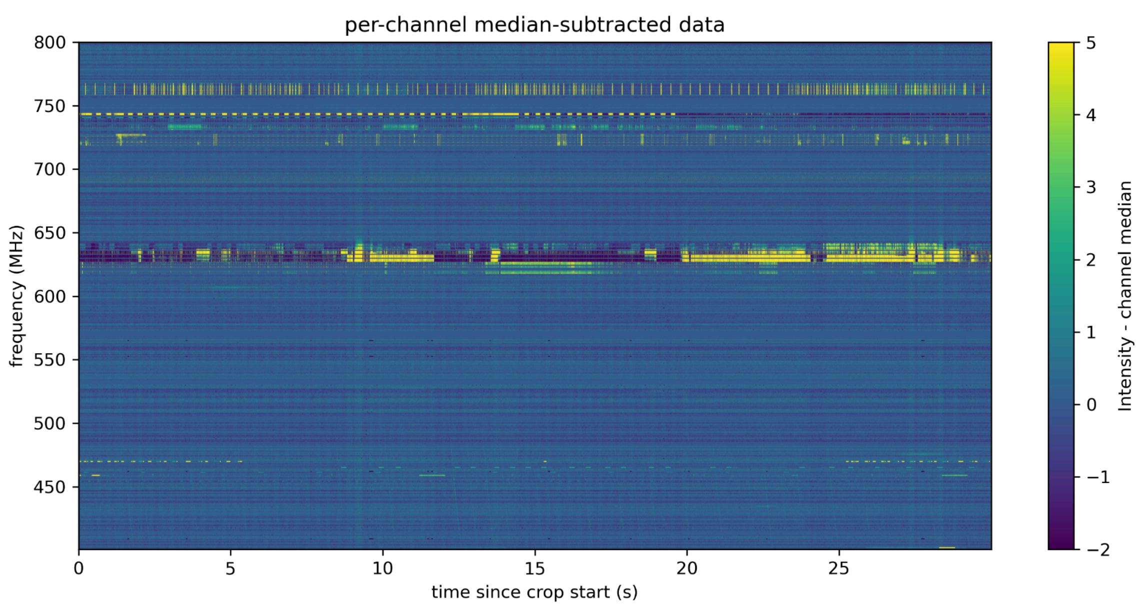

Fig 1. Frequency over cropped time window from 5 to 35 seconds with the median channel intensity subtracted.

A two-dimensional mask was then applied to the remaining frequency and time channels with 0.75 sigma deviation or higher from the median intensity and masking was manually applied to known bad frequency channels. The 0.75 sigma value was determined based on a 1D time series vs channel power diagnostic.

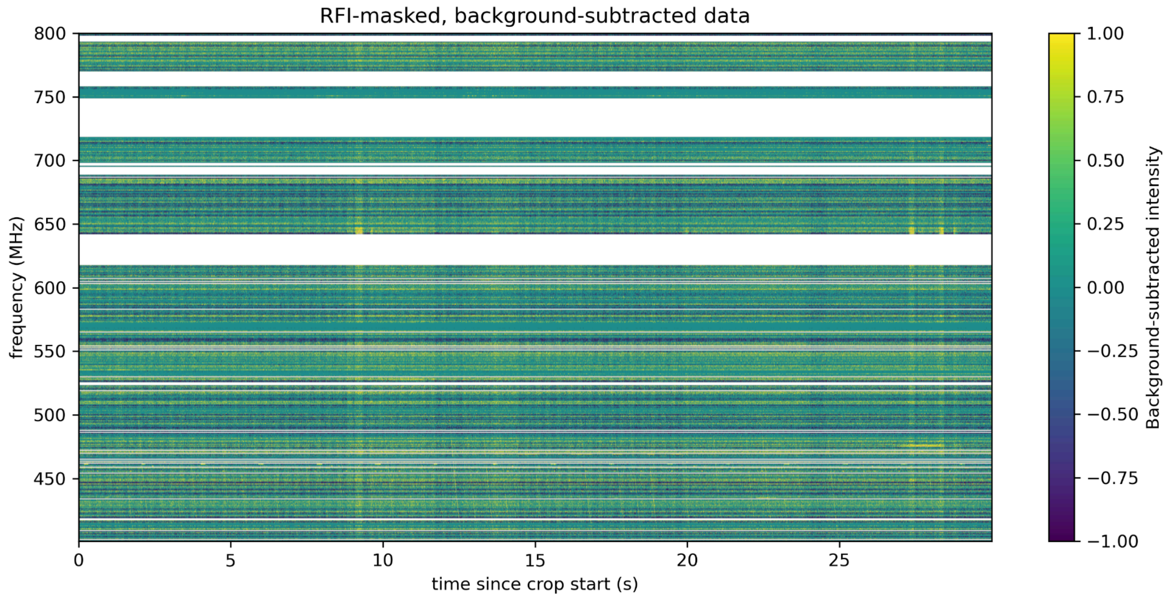

Fig 2. Frequency over cropped time window from 5 to 35 seconds with the median channel intensity subtracted and channels with over 0.75 median deviation masked.

After the masking, a dispersion measure diagnostic was created by dedispersing the masked spectrum over a range of DMs (from 0 to 100), collapsing each dispersed array into a 1D time series, and measuring the resulting SNR as a function of DM.

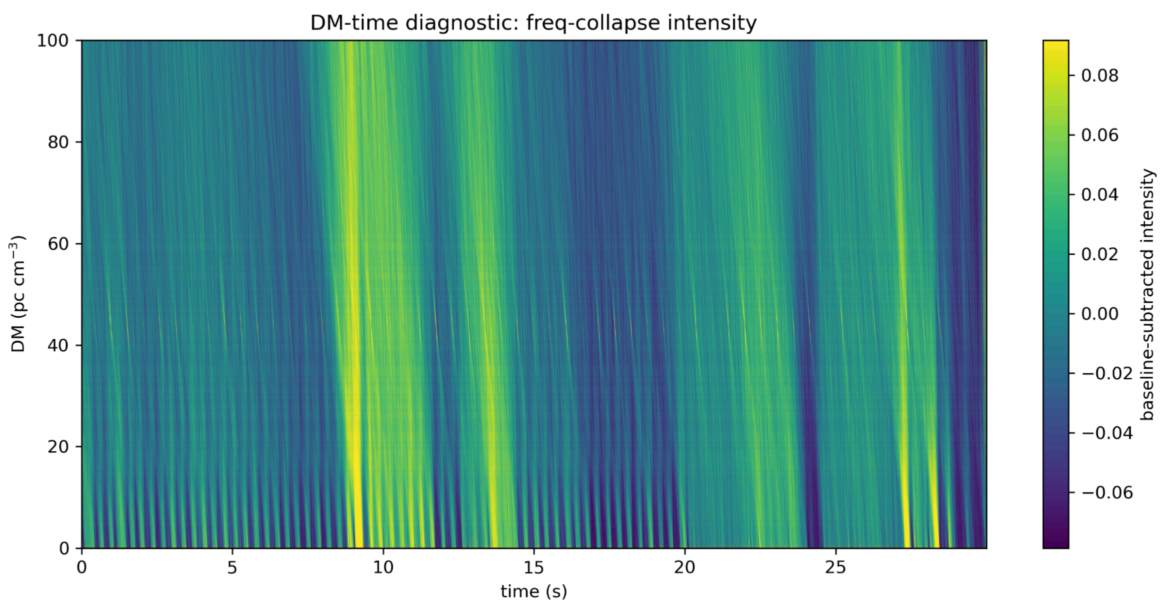

Fig 3. Diagnostic plot of SNR as a function of DM over time window.

The DM with the best SNR was used to dedisperse the time series, which was then transformed into spin-frequency space using a fourier transform (FFT). Another median subtraction was performed to normalize the red noise before the harmonics were stacked to amplify the signals. The DM versus the spin frequency was plotted as in terms of the logarithmic stacked power. The birdies, bad spin channels, were determined as having a high power at DM=0, and were masked.

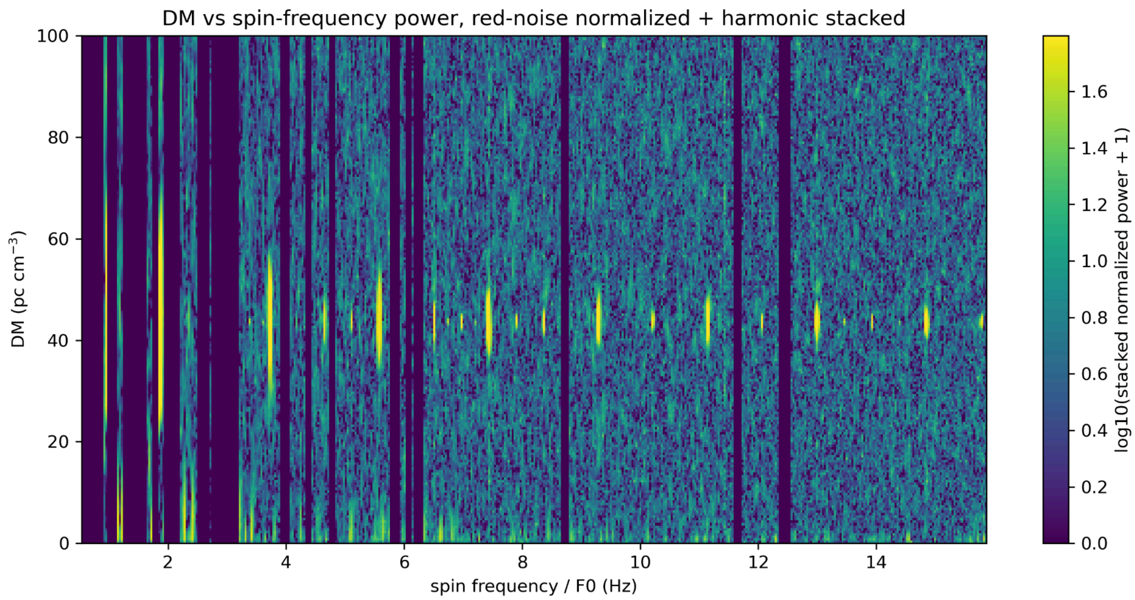

Fig 4. Power as a function of DM over stacked harmonic spin frequencies with red noise normalized and birdies masked.

Next, a function was created to search for harmonic clusters and then identify their maxima. Using the pandas library, a data table of fundamental frequency candidates was created and ranked based on SNR. For each candidate, this table included the weighted average fundamental frequency, weighted average DM, the cluster SNR, the number of harmonics, and which harmonics were detected. The table was set to save as a csv in the same folder as all plots. Using the F_0 value associated with the highest SNR, the signal was then folded to create a pulse profile in order to extract a value for its phase.

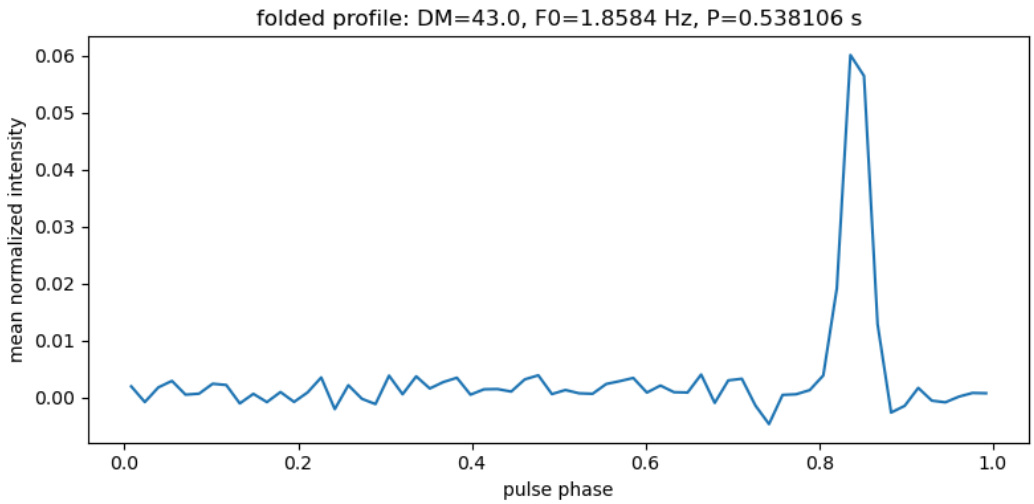

Fig 5. Pulse folded profile using the best SNR weighted fundamental frequency of 1.8584 Hz

Results

Each extracted value was based on having the greatest SNR. In this way, the fundamental frequency was found to be $F_0 = 1.8584\, Hz$, making the corresponding period $P = 0.5381\, s$.

The dispersion measure was found to be $DM = 43.0\, pc\, cm^{-3}$. The phase was found to be $\Phi \approx 0.83\, cycles = 0.447\, s$. Based on these values, and keeping in mind the right ascension and declination found in the conventional naming of the dataset, the pulsar was identified as B2217+47 using the data from the ANTF pulsar catalog.

Discussion

Due to the timescale of this project, it wasn't possible to meaningfully estimate the uncertainty, though many measures were taken to mitigate as much error as possible. Given the brightness of this pulsar, the fundamental frequency extracted matched that listed in the pulsar catalog to $\pm\ 0.001\, Hz$, and the pulse profile phase matched to $\pm\ 0.1\, cycle$, meaning $\pm\ 0.005s$. This error value is not insignificant, especially as the sampling rate was 0.00196s. One unutilized way to mitigate this error further would've been to use more than 30s for the analysis, perhaps 2-5 minutes if not the full 16 minutes.

In the future, this program still needs to be more sensitively tuned to fainter pulsar signals, and does not currently accommodate for the possibility of there being more than one pulsar signal. In addition, if there were more than one pulsar signal, it would be beneficial to create a ranked list of all pulsar candidates with their extracted statistics for DM, F_0, Φ, and harmonics number.

Conclusion

The objective of this project, to create a data analysis pipeline which would automatically detect a pulsar signal, strengthen it with stacking, and fold to create a pulse profile, was met with the condition that the pulsar must be of a certain brightness. Given the timeline of this project, it was not possible to make a meaningful uncertainty estimate, and all error values were based in comparison to the accepted metrics for B2217+47 in the ATNF pulsar catalogue. The most immediate next step for this project would be to diagnose a more significant uncertainty measurement. One way to do this would be to input a simulated pulsar candidate in order to validate the resulting measurements. It is also necessary to improve the code's sensitivity to fainter pulsars and to create a pulsar candidate ranking system in the event that there are multiple signals found.

References

Andrade, C., Boyle, P. J., Brar, C., Cassity, A., Crowter, K., Cubranic, D., Denney, A. K., Dong, F. A., Fonseca, E., Kaspi, V. M., Kumar, A., Künkel, L., L'Argent, M., Lang, D., Main, R. A., Masui, K. W., Mate, S., Mena-Parra, J., Meyers, B. W., … Zegmott, T. J. (2025). Chime all-sky multiday Pulsar Stacking Search (CHAMPSS): System overview and first discoveries. The Astrophysical Journal, 990(1), 50. https://doi.org/10.3847/1538-4357/adeb51